

CSE402 Assignment #1: Matrix Calculus and Naive Bayes Classification

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1. Matrix Derivatives in Jacobian Formulation (22 points)

Suppose that $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is a column vector, defined as follows:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad (1)$$

We now define a function of vector $\mathbf{x}$, i.e., $\mathbf{y} = \mathbf{f}(\mathbf{x}) \in \mathbb{R}^{m \times 1}$. The Jacobian matrix, $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}$ – that is, the partial derivative of $\mathbf{y}$ with respect to $\mathbf{x}$ – is given as follows:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \cdots & \frac{\partial y_m}{\partial x_n} \end{pmatrix} \quad (2)$$

In addition, when $\mathbf{X} \in \mathbb{R}^{n \times m}$ is a parameter matrix and $y = f(\mathbf{X})$ is a scalar function of $\mathbf{X}$, the Jacobian matrix $\frac{\partial y}{\partial \mathbf{X}} \in \mathbb{R}^{m \times n}$ is defined as follows:

$$\frac{\partial y}{\partial \mathbf{X}} = \begin{pmatrix} \frac{\partial y}{\partial x_{11}} & \frac{\partial y}{\partial x_{21}} & \cdots & \frac{\partial y}{\partial x_{n1}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y}{\partial x_{1m}} & \frac{\partial y}{\partial x_{2m}} & \cdots & \frac{\partial y}{\partial x_{nm}} \end{pmatrix} \quad (3)$$

(a) (3 points) Suppose that $\mathbf{z} \in \mathbb{R}^k$, $\mathbf{y} \in \mathbb{R}^n$, and $\mathbf{x} \in \mathbb{R}^m$ are given as follows:

$$\begin{aligned} \mathbf{z} &= f(\mathbf{y}) \\ \mathbf{y} &= g(\mathbf{x}) \end{aligned} \quad (4)$$

Prove that the following chain rule holds:

$$\frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \frac{\partial \mathbf{z}}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \quad (5)$$

Note: Please specify the shapes of all matrices when they appear in the derivation.

$$\mathbf{x} \in \mathbb{R}^m, \quad \mathbf{y} = g(\mathbf{x}) \in \mathbb{R}^n, \quad \mathbf{z} = f(\mathbf{y}) \in \mathbb{R}^k$$

$$\frac{d\mathbf{z}}{d\mathbf{x}} \in \mathbb{R}^{k \times m}, \quad \frac{d\mathbf{z}}{d\mathbf{y}} \in \mathbb{R}^{k \times n}, \quad \frac{d\mathbf{y}}{d\mathbf{x}} \in \mathbb{R}^{n \times m}$$

$$\frac{d\mathbf{z}}{d\mathbf{x}} = \begin{pmatrix} \frac{dz}{dx_1} & \cdots & \frac{dz}{dx_m} \\ \vdots & \ddots & \vdots \\ \frac{dz_k}{dx_1} & \cdots & \frac{dz_k}{dx_m} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\partial z_1}{\partial y_1} \cdot \frac{\partial y_1}{\partial x_1} + \dots + \frac{\partial z_1}{\partial y_n} \cdot \frac{\partial y_n}{\partial x_1} & \dots & \frac{\partial z_1}{\partial y_1} \cdot \frac{\partial y_1}{\partial x_m} + \dots + \frac{\partial z_1}{\partial y_n} \cdot \frac{\partial y_n}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_u}{\partial y_1} \cdot \frac{\partial y_1}{\partial x_1} + \dots + \frac{\partial z_u}{\partial y_n} \cdot \frac{\partial y_n}{\partial x_1} & \dots & \frac{\partial z_u}{\partial y_1} \cdot \frac{\partial y_1}{\partial x_m} + \dots + \frac{\partial z_u}{\partial y_n} \cdot \frac{\partial y_n}{\partial x_m} \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{l=1}^n \frac{\partial z_1}{\partial y_l} \frac{\partial y_l}{\partial x_1} & \dots & \sum_{l=1}^n \frac{\partial z_1}{\partial y_l} \frac{\partial y_l}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \sum_{l=1}^n \frac{\partial z_u}{\partial y_l} \frac{\partial y_l}{\partial x_1} & \dots & \sum_{l=1}^n \frac{\partial z_u}{\partial y_l} \frac{\partial y_l}{\partial x_m} \end{pmatrix}$$

matrix product

$$= \begin{pmatrix} \frac{\partial z_1}{\partial y_1} & \dots & \frac{\partial z_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_u}{\partial y_1} & \dots & \frac{\partial z_u}{\partial y_n} \end{pmatrix} \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \dots & \frac{\partial y_n}{\partial x_m} \end{pmatrix}$$

$$= \frac{\partial z}{\partial y} \frac{\partial y}{\partial x}$$

$$\text{shape: } (k \times n)(n \times m) = (k \times m)$$

$$\text{So, } \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} \frac{\partial y}{\partial x}$$

(b) (3 points) Prove that if $\mathbf{y} = \mathbf{Ax}$, where $\mathbf{y} \in \mathbb{R}^{m \times 1}$, $\mathbf{x} \in \mathbb{R}^{n \times 1}$, and $\mathbf{A} \in \mathbb{R}^{m \times n}$, then the Jacobian matrix of $\mathbf{y}$ with respect to $\mathbf{x}$ is given by:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbf{A} \quad (6)$$

$$\mathbf{y} = \mathbf{Ax}$$

$$y_i = \sum_{k=1}^n A_{ik} x_k$$

$$\text{then } \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \dots & \frac{\partial y_m}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial \left(\sum_{k=1}^n A_{1k} x_k \right)}{\partial x_1} & \dots & \frac{\partial \left(\sum_{k=1}^n A_{1k} x_k \right)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \left(\sum_{k=1}^n A_{mk} x_k \right)}{\partial x_1} & \dots & \frac{\partial \left(\sum_{k=1}^n A_{mk} x_k \right)}{\partial x_n} \end{pmatrix}$$

$$\text{Because } \frac{\partial (A_{ij} x_j)}{\partial x_k} = 0 \text{ for } j \neq k,$$

$$= \begin{pmatrix} \frac{\partial (A_{11} x_1)}{\partial x_1} & \dots & \frac{\partial (A_{1n} x_n)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial (A_{m1} x_1)}{\partial x_1} & \dots & \frac{\partial (A_{mn} x_n)}{\partial x_n} \end{pmatrix} = \begin{pmatrix} A_{11} & \dots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mn} \end{pmatrix} = \mathbf{A}$$

(c) (3 points) Prove that if $\mathbf{y} = f(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^{n \times 1}$ and f is an elementwise function, then the Jacobian matrix of $\mathbf{y}$ with respect to $\mathbf{x}$ is given by:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \text{diag}(f'(\mathbf{x})) \quad (7)$$

Let $x \in \mathbb{R}^{n \times 1}$ and $y = f(x) \in \mathbb{R}^{n \times 1}$.

$$y = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix}$$

$$\frac{\partial y}{\partial x} = \left(\frac{\partial y_i}{\partial x_j} \right) \in \mathbb{R}^{n \times n}$$

Since f is elementwise, $y_i = f(x_i)$

$$\frac{\partial y_i}{\partial x_j} = \begin{cases} f'(x_i), & i=j \\ 0, & i \neq j \end{cases}$$

So, Jacobian becomes a diagonal matrix.

$$\frac{\partial y}{\partial x} = \begin{pmatrix} f'(x_1) & 0 & \dots & 0 \\ 0 & f'(x_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & f'(x_n) \end{pmatrix} = \text{diag}(f'(x))$$

$$\text{So, } \frac{\partial y}{\partial x} = \text{diag}(f'(x))$$

(d) (3 points) If $\mathbf{z} = f(\mathbf{U}g(\mathbf{W}\mathbf{x}))$, where $\mathbf{z} \in \mathbb{R}^{k \times 1}$, $\mathbf{U} \in \mathbb{R}^{k \times n}$, $\mathbf{W} \in \mathbb{R}^{n \times m}$, $\mathbf{x} \in \mathbb{R}^{m \times 1}$, and f and g are elementwise functions, what is the Jacobian matrix $\frac{\partial \mathbf{z}}{\partial \mathbf{x}}$?

$$\frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \quad (8)$$

$$\frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \frac{\partial (f(\mathbf{U}g(\mathbf{W}\mathbf{x})))}{\partial \mathbf{x}} = \frac{\partial (f(\mathbf{U}g(\mathbf{W}\mathbf{x})))}{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))} \cdot \frac{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))}{\partial \mathbf{x}}$$

$\downarrow$ $(k \times m)$ $\downarrow$ $(k \times n)$ $\downarrow$ $(n \times m)$

$$= \frac{\partial (f(\mathbf{U}g(\mathbf{W}\mathbf{x})))}{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))} \cdot \frac{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))}{\partial (g(\mathbf{W}\mathbf{x}))} \cdot \frac{\partial (g(\mathbf{W}\mathbf{x}))}{\partial \mathbf{x}}$$

$\downarrow$ $(k \times n)$ $\downarrow$ $(n \times m)$

$$= \frac{\partial (f(\mathbf{U}g(\mathbf{W}\mathbf{x})))}{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))} \cdot \frac{\partial (\mathbf{U}g(\mathbf{W}\mathbf{x}))}{\partial (g(\mathbf{W}\mathbf{x}))} \cdot \frac{\partial (g(\mathbf{W}\mathbf{x}))}{\partial (\mathbf{W}\mathbf{x})} \cdot \frac{\partial (\mathbf{W}\mathbf{x})}{\partial \mathbf{x}}$$

$\downarrow$ $(k \times n)$ $\downarrow$ $(n \times n)$ $\downarrow$ $(n \times m)$

$$= \text{diag}(f'(\mathbf{U}g(\mathbf{W}\mathbf{x}))) \cdot \mathbf{U} \cdot \text{diag}(g'(\mathbf{W}\mathbf{x})) \cdot \mathbf{W}$$

$(k \times k)$ $(k \times n)$ $(n \times n)$ $(n \times m)$

(e) (5 points) Prove that if $J = h(\mathbf{z})$ and $\mathbf{z} = \mathbf{W}\mathbf{x}$, where $\mathbf{z} \in \mathbb{R}^{m \times 1}$ and $\mathbf{x} \in \mathbb{R}^{n \times 1}$, then the Jacobian matrix of J with respect to $\mathbf{W}$ is given by:

$$\frac{\partial J}{\partial \mathbf{W}} = \mathbf{x} \frac{\partial J}{\partial \mathbf{z}} \quad (9)$$

Each element, $z_i = \sum_{j=1}^n w_{ij} x_j$

$$\Rightarrow \frac{\partial z_i}{\partial w_{pq}} = \begin{cases} x_q, & i=p \\ 0, & i \neq p \end{cases}$$

Apply chain rule, $\frac{\partial J}{\partial w_{pq}} = \sum_{i=1}^m \frac{\partial J}{\partial z_i} \frac{\partial z_i}{\partial w_{pq}}$.

Using this, $\frac{\partial J}{\partial w_{pq}} = \frac{\partial J}{\partial z_p} x_q$

By the definition of Jacobian with respect to a matrix,

$$\left(\frac{\partial J}{\partial \mathbf{W}} \right)_{q,p} = \frac{\partial J}{\partial w_{pq}} = x_q \frac{\partial J}{\partial z_p}$$

$$\frac{\partial J}{\partial \mathbf{W}} = \begin{pmatrix} x_1 \frac{\partial J}{\partial z_1} & \cdots & x_1 \frac{\partial J}{\partial z_m} \\ \vdots & \ddots & \vdots \\ x_n \frac{\partial J}{\partial z_1} & \cdots & x_n \frac{\partial J}{\partial z_m} \end{pmatrix}$$

So, $\frac{\partial J}{\partial \mathbf{W}} = \mathbf{x} \frac{\partial J}{\partial \mathbf{z}}$.

$$\mathbf{x} \in \mathbb{R}^{n \times 1}, \quad \frac{\partial J}{\partial \mathbf{z}} \in \mathbb{R}^{1 \times m}, \quad \mathbf{x} \frac{\partial J}{\partial \mathbf{z}} \in \mathbb{R}^{n \times m}$$

(f) (5 points) Prove that if the cross-entropy loss CE is expressed as:

$$\begin{aligned} J &= CE(\mathbf{y}, \hat{\mathbf{y}}) = -\mathbf{y}^T \log \hat{\mathbf{y}} \\ \hat{\mathbf{y}} &= \text{softmax}(\mathbf{z}) \end{aligned} \quad (10)$$

where $\mathbf{y} \in \mathbb{R}^{k \times 1}$ is the one-hot label vector and $\mathbf{z} \in \mathbb{R}^{k \times 1}$ are the logits, then the Jacobian matrix of the cross-entropy loss J with respect to the logits $\mathbf{z}$ is given by:

$$\frac{\partial J}{\partial \mathbf{z}} = (\hat{\mathbf{y}} - \mathbf{y})^T \quad (11)$$

For each i , $\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$.

$$\log \hat{y}_i = z_i - \log \left(\sum_{j=1}^n e^{z_j} \right) = z_i - 1 \cdot \log \left(\sum_{j=1}^n e^{z_j} \right)$$

$$\text{So, } J = -\mathbf{y}^T \log \hat{\mathbf{y}} = -\mathbf{y}^T \mathbf{z} + \mathbf{y}^T \cdot 1 \cdot \log \left(\sum_{j=1}^n e^{z_j} \right)$$

Since $\mathbf{y}$ is one-hot, $\mathbf{y}^T \mathbf{1} = 1$.

$$\text{Thus, } J = -\mathbf{y}^T \mathbf{z} + \log \left(\sum_{j=1}^n e^{z_j} \right)$$

Using Jacobian formulation,

$$\frac{\partial J}{\partial \mathbf{z}} = -\mathbf{y}^T + \frac{\partial}{\partial \mathbf{z}} \log \left(\sum_{j=1}^n e^{z_j} \right)$$

$$\text{Now, } \frac{\partial}{\partial \mathbf{z}} \log \left(\sum_{j=1}^n e^{z_j} \right) = \frac{1}{\sum_{j=1}^n e^{z_j}} (e^{z_1} \ e^{z_2} \ \dots \ e^{z_n}) = \hat{\mathbf{y}}^T$$

$$\text{So, } \frac{\partial J}{\partial \mathbf{z}} = -\mathbf{y}^T + \hat{\mathbf{y}}^T = (\hat{\mathbf{y}} - \mathbf{y})^T$$

2. Naive Bayesian Text Classifier (24 points)

In text classification, a text $\mathbf{x}$ is treated as a sequence of words, i.e., $\mathbf{x} = (w_1, \dots, w_n)$.

The Naïve Bayes model for text classification assumes that words are generated independently, and thus uses the posterior probability as follows:

$$P(y = k | \mathbf{x}) = \frac{\prod_{i=1}^n P(w_i | y = k) P(y = k)}{P(\mathbf{x})} \quad (12)$$

- (a) (3 points) Consider the frequency-based estimator for the word generative probability $P(w_i | y = k)$, given by:

$$P(w_i | y = k) \approx \frac{c(w_i, y = k)}{\sum_j c(w_j, y = k)} \quad (13)$$

If a word w_i never appears in training texts of class k , where $c(w_i, y = k)$ is the count (or frequency) of word w_i appearing in texts belonging to class k , ($y = k$) in the dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, and $\sum_j c(w_j, y = k)$ is the **total count** of all words appearing in texts belonging to class k , ($y = k$).

then $P(w_i, y = k) = 0$.
What is the problem with the frequency-based estimator?

So $P(w_i, y = k) = 0$.

Since Naive Bayes multiplies the word probabilities,

$$P(y = k | \mathbf{x}) \propto \prod_{i=1}^n P(w_i | y = k) P(y = k).$$

If any word in the text has $P(w_i | y = k) = 0$,

then the whole product becomes zero, $P(y = k | \mathbf{x}) = 0$

This is called the zero-probability problem,

and it can be solved by smoothing method such as

Laplace smoothing.

(b) (21 points) We model the word generative probability $P(w_i | y = k)$ using word embeddings as follows:

$$P(w_i | y = k) = \text{softmax}_i(\mathbf{v}_i^\top \mathbf{c}_k) = \frac{\exp(\mathbf{v}_i^\top \mathbf{c}_k)}{\sum_{j \in \mathcal{V}} \exp(\mathbf{v}_j^\top \mathbf{c}_k)} \quad (14)$$

Consider the negative log-likelihood as the objective (loss) function for the Naïve Bayes model, defined as follows:

$$\begin{aligned} \text{Loss} &= - \sum_{(\mathbf{x}, k) \in \mathcal{D}} \log P(\mathbf{x}, y = k) \\ &= - \sum_{(\mathbf{x}, k) \in \mathcal{D}} \log P(\mathbf{x} | y = k) P(y = k) \\ &= - \sum_{(\mathbf{x}, k) \in \mathcal{D}} \log \left(\prod_{w_i \in \mathbf{x}} P(w_i | y = k) \right) P(y = k) \\ &= - \underbrace{\sum_{(\mathbf{x}, k) \in \mathcal{D}} \sum_{w_i \in \mathbf{x}} \log P(w_i | y = k)}_{\text{conditional likelihood part}} - \underbrace{\sum_{k=1}^K N_k \log P(y = k)}_{\text{prior part}} \end{aligned} \quad (15)$$

i. (8 points) Suppose the word embedding matrix is given by:

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_{|\mathcal{V}|}^\top \end{bmatrix} \quad (16)$$

Derive the Jacobian matrix of the loss Loss in Eq. (15) with respect to $\mathbf{V}$, expressed as:

$$\frac{\partial \text{Loss}}{\partial \mathbf{V}} = \quad (17)$$

For each class k ,

$$\begin{aligned} \text{define } \mathbf{z} &= \mathbf{V} \mathbf{c}_k \in \mathbb{R}^{|\mathcal{V}| \times 1}, \quad P_k = \text{softmax}(\mathbf{z}) = \text{softmax}(\mathbf{V} \mathbf{c}_k), \\ & \left(\mathbf{V} \in \mathbb{R}^{|\mathcal{V}| \times d}, \quad \mathbf{c}_k \in \mathbb{R}^{d \times 1} \right) \end{aligned}$$

$$\text{From Eq. (15), } \text{Loss} = - \sum_{(\mathbf{x}, k) \in \mathcal{D}} \sum_{w_i \in \mathbf{x}} \log P(w_i | y = k) - \sum_{k=1}^K N_k \log P(y = k).$$

The prior term $\sum_{k=1}^K N_k \log P(y = k)$ does not depend on $\mathbf{V}$,

so its derivative with respect to $\mathbf{V}$ is zero.

Thus, it is enough to differentiate $-\log P(w_i | y = k)$.

Let $e_{w_i} \in \mathbb{R}^{|V| \times 1}$ be one-hot vector for word w_i ,

$$-\log P(w_i | y=k) = -e_{w_i}^T \log \text{softmax}(z).$$

Using cross-entropy derivative

← by 1 (f)

$$\frac{\partial J}{\partial z} = (\text{softmax}(z) - e_{w_i})^T = (p_k - e_{w_i})^T$$

Since $z = V c_u$,

$$\frac{\partial J}{\partial V} = c_u \frac{\partial J}{\partial z} = c_u (p_k - e_{w_i})^T$$

← by 1 (e)

$$\text{So, } \frac{\partial \text{Loss}}{\partial V} = \sum_{(x,k) \in D} \sum_{w_i \in x} c_k (\text{softmax}(V c_u) - e_{w_i})^T$$

$$\frac{\partial \text{Loss}}{\partial V} = \sum_{(x,k) \in D} \sum_{w_i \in x} c_k (P(\cdot | y=k) - e_{w_i})^T$$

where $P(\cdot | y=k) = \text{softmax}(V c_k) \in \mathbb{R}^{|V| \times 1}$.

ii. (5 points) Derive the Jacobian matrix of the loss $Loss$ in Eq. (15) with respect to $\mathbf{c}_k$, expressed as:

$$\frac{\partial Loss}{\partial \mathbf{c}_k} = \quad (18)$$

$$\text{Let } Vc_u \in \mathbb{R}^{|V| \times 1}, \quad p_u = \text{softmax}(Vc_u) \in \mathbb{R}^{|V| \times 1}$$

$$\text{where } V \in \mathbb{R}^{|V| \times d}, \quad c_u \in \mathbb{R}^{d \times 1}.$$

$$\text{From Eq. (15), } Loss = - \sum_{(x,y) \in D} \sum_{w_i \in x} \log P(w_i|y) - \sum_{r=1}^k N_r \log P(y=r).$$

Since the prior term does not depend on c_u , its derivative with respect to c_u is zero.

Also, only the examples with class label u contribute to $\frac{\partial Loss}{\partial c_u}$.

$$\frac{\partial Loss}{\partial c_u} = \sum_{(x,u) \in D} \sum_{w_i \in x} \frac{\partial}{\partial c_u} (-\log P(w_i|y=u)).$$

Let $e_{w_i} \in \mathbb{R}^{|V| \times 1}$ be the one-hot vector for word w_i .

$$-\log P(w_i|y=u) = -e_{w_i}^T \log(\text{softmax}(z))$$

$$\text{By 1 (f), } \frac{\partial J}{\partial z} = (\text{softmax}(z) - e_{w_i})^T = (p_u - e_{w_i})^T.$$

Since $z = Vc_u$, we have $\frac{\partial z}{\partial c_u} = V$.

$$\text{By chain rule, } \frac{\partial J}{\partial c_u} = \frac{\partial J}{\partial z} \cdot \frac{\partial z}{\partial c_u} = (p_u - e_{w_i})^T V$$

$$\text{So, } \frac{\partial Loss}{\partial c_u} = \sum_{(x,u) \in D} \sum_{w_i \in x} (\text{softmax}(Vc_u) - e_{w_i})^T V,$$

$$\frac{\partial Loss}{\partial c_u} = \sum_{(x,u) \in D} \sum_{w_i \in x} (P(\cdot|y=u) - e_{w_i})^T V.$$

- iii. (8 points) Note that the denominator in Eq. (14) involves a summation of exponentiated logits over all words. If you choose to use a negative sampling method, as in word2vec (i.e., skip-gram with negative sampling), how would you revise the loss function? Here, it is not necessary to preserve the precise form of the joint likelihood.

Using negative sampling, we replace softmax probability with a binary classification objective.

For a positive word w_i and a set of negative samples $N(i)$,

$$P(w_i | y=k) \approx \sigma(v_i^T c_k) \prod_{j \in N(i)} \sigma(-v_j^T c_k).$$

Therefore,

$$\begin{aligned} L_{\text{loss}} = & - \sum_{(x,y) \in D} \sum_{w_i \in x} \left(\log \sigma(v_i^T c_y) + \sum_{j \in N(i)} \log \sigma(-v_j^T c_y) \right) \\ & - \sum_{k=1}^K N_k \log P(y=k). \end{aligned}$$